| Version:1 |
|---|

Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

# ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

## OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

# SPINAL ANAESTHESIA FOR CAESAREAN SECTION

| | | | |
|---|---|---|---|
| **Date to be reviewed:** | 1st April 2029 | **No of pages:** | 7 |
| **Author(s):** | Dr Gareth Roberts | **Author(s) title:** | Consultant Anaesthetist |
| **Responsible dept / director:** | Department of Anaesthesia, BCUHB (West)<br>Dr Lisa Handcock Clinical Lead of Anaesthesia, BCUHB (West) | | |
| **Approved by:** | Department of Anaesthesia, BCUHB (West) | | |
| **Date approved:** | | | |
| **Endorsement by:** | Department of Anaesthesia, BCUHB (West) | | |
| **Date endorsed:** | | | |
| **Date activated (live):** | 1st April 2026 | | |

| | |
|---|---|
| **Date EQIA completed:** | |
| **Documents to be read alongside this policy:** | None |
| **Review** | **Purpose of Issue/Description of current changes:**<br>New obstetric anaesthesia written control document for Betsi Cadwaladr University Health Board (West) |

| | | | | | |
|---|---|---|---|---|---|
| **First operational:** | January 2013 | | | | |
| **Previously reviewed:** | Date: 30/1/13 | Date: 1/10/18 | Date: 23/07/21 | Date: 1/5/24 | Date: 11/3/26 |
| **Changes made yes/no:** | Yes | Yes | Yes | Yes | Yes |

PROPRIETARY INFORMATION
This document contains proprietary information belonging to the Betsi Cadwaladr University Health Board. Do not produce all or any part of this document without written permission from the BCU

## Contents

1. Introduction and purpose ……3
2. Scope……..……3
3. Points to consider……...3
4. References……..6


Paper copies of this document should be kept to a minimum and checks made with the electronic version to ensure the version to hand is the most recent.

## 1. Introduction and Purpose

Spinal anaesthesia has been the commonest type of anaesthesia given for both elective and emergency Caesarean Section in the UK since the mid 1980's. It is safer than general anaesthesia in obstetrics since it avoids airway associated morbidity and mortality. However it is associated with risks such as post dural puncture headache (1:100) and nerve damage (1:24,000 risk of permanent nerve damage at 6 months).

## 2. Scope

This guideline is intended as a reference for anaesthetic trainees seeking a recommended technique for spinal anaesthesia in obstetrics.

## 3. Points to consider

3.1 Patient Preparation

- Full preoperative assessment of the patient as for a GA
- Ensure no contraindications to spinal block (patient refusal, thrombocytopaenia (platelet count < 80), coagulopathy (INR >1.4), anticoagulation therapy, local sepsis, systemic sepsis or uncorrected hypovolaemia
- Discussion with the patient of:

a) Benefits of a spinal block vs GA:

i. Safer due to less risk of aspiration of gastric contents and failed intubation

ii. Patient able to be awake to witness the birth with partner

iii. Earlier breast feeding

iv. Better early post operative analgesia (if long acting intrathecal opiates e.g. diamorphine used)

v. No hangover effect as with a GA

b) Risks of a spinal block :

i. Post dural puncture headache (1:100)

ii. Nausea and vomiting

iii. Failure (including need for intraoperative GA conversion)

iv. Nerve damage (Temporary = 1:1,000,
Permanent = 1:24,000 incidence at 6 months)

c) Spinal block procedure

d) Explanation of what sensations she will feel intraoperatively

- Obtain verbal consent for spinal block and document in the patient casenotes.

- Discussion with the patient about postoperative analgesia ( Regular Paracetamol 1 gram P.O. 6 hourly, regular Ibuprofen 600mg P.O. 6 hourly, & Oramorph 25mg 2 hourly PRN for Caesarean Sections, if no contraindications to these drugs)

- For elective Caesarean Sections, patients should be fasted for 6 hours but allowed sips of water until theatre (max 30ml/hour for the last 2 hours prior to theatre)

- Antacid prescription : Omeprazole 40mg at 22:00 (previous day), Omeprazole 40mg at 07:00 (on day of surgery)

3.2 Equipment and Staff

- Check all anaesthetic equipment as for GA

- Draw up drugs as for GA and store in fridge

- Ensure metaraminol, ephedrine and atropine at hand

- Ensure ODP is present prior to commencing spinal block procedure

3.3 Technique

- Attach full routine monitoring (ECG, NIBP, Pulse Oximetry)

- Site a **16G** (or 14G) iv cannula

- Preload with **1,000 ml Hartmanns solution** iv (use 500ml crystalloid preload in severe pre-eclampsia)

- Place patient in **Sitting** or lateral position

- Use full aseptic technique :

  a) Gown, mask and gloves

  b) Spray the patient's back twice from the mid thoracic area to the upper natal cleft, and over to the flanks using **Derma Spray** (0.5% chlorhexidine in 70% alcohol). Allow the alcohol to dry each time.

- Skin and ligament infiltration with 1% Lignocaine at **L3/4** or L4/5 interspace

- Use **25G Pencil Point** spinal needle with introducer (90mm needle (30mm introducer) for thin patients or 103mm (38mm introducer) for larger patients

- Obtain free flowing CSF flashback into hub on removing stylet

- Administer **2.5mls 0.5% Hyperbaric bupivacaine + 300 micrograms diamorphine* (or alternatively 15 - 20 micrograms of fentanyl)**

- Remove needle and apply plaster to injection site

- Place patient supine with 15 degree left lateral tilt (to prevent aorto-caval compression) immediately following removal of spinal needle. To achieve this the operating table will need to be placed in head down position when the patient is positioned on the Oxford HELP pillow

- Give patient 4l oxygen via Hudson facemask for predelivery period if maternal oxygen saturation is below 95% or significant foetal distress is present on CTG

- Perform routine haemodynamic measurements (pulse, NIBP) every 1 minute for 5 minutes, then every 2 minutes for the next 5 minutes. Subsequently take readings every 3-5 minutes if the patient is cardiovascularly stable, otherwise more often.

- Allow midwives to place urinary catheter

- If the patient develops a tachycardia following spinal anaesthesia (usually due to vasodilation) administer metaraminol (0.5mg boluses) and increase the rate of intravenous fluid administration, to prevent the development of hypotension. Alternatively, commence a metaraminol infusion (0.5mg/ml) at 20ml/hour immediately following spinal anaesthesia insertion, and titrate the dose to maintain a systolic Bp >100mmHg. Aim to wean off the metaraminol infusion following delivery.

- If subsequently the systolic BP drops below 100 mmHg (or below 20% of baseline BP) check that the amount of left lateral tilt is adequate (ie. true 15 degrees). Also, increase the rate of intravenous fluid administered, and give additional vasopressor drug. Choose metaraminol if HR > 60/min (0.5mg boluses) or ephedrine if HR < 60/min (6mg boluses).

- Assess level of sensory block (temperature) using the Coolstick or Ethyl Chloride spray

- Assess the degree of motor block by asking the patient to lift her legs

- Commence surgery only when sensory level of block extends uniformly from S5 to either of the following levels :

  i T4 when assessed by cold sensation

  ii T5 when assessed by light touch

  Significant motor blockade should also be present

- Prior to commencement of surgery administer administer iv antibiotic (Cefuroxime 1.5g & Metronidazole 500mg; or if the patient has a previous history of anaphylaxis to penicillin then give Clindamycin 900mg & Gentamycin 240mg iv (or 160mg if body weight <60Kg))

- After delivery of the baby :

    i. Administer 5iu bolus dose of syntocinon

    ii. **If requested by surgeon** (for atonic uterus) set up infusion of 40iu syntocinon in 500mls 0.9% Saline and run at a rate of 125 mls / hr

- Administer paracetamol 1 gram iv

- Administer Dexamethasone 0.1mg/Kg iv

- Administer Ondansetron 4 mg/Kg iv

- Administer sodium diclofenac 100mg suppository PR at end of procedure (if no contraindications)

- Proceed to the recovery area with the patient. Do not leave until the patient is fully conscious, pain free and has stable vital signs

3.4 Troubleshooting

- Keep in close verbal contact with patient throughout surgery to explain the various sensations felt

- If significant discomfort and especially sharp pain is felt during surgery ask the surgeon to stop operating and give supplementary rapid onset intravenous opiates e.g. fentanyl (25 – 50 microgram boluses) or alfentanil boluses before recommencing surgery. Also, consider administering Entonox. In cases of sharp pain or severe discomfort which does not resolve despite administering 150 micrograms of fentanyl intravenously, then strongly recommend a general anaesthetic to the patient, and proceed if she consents. If she declines a GA then document this.

- If systolic BP is below 100mmHg (or below 20% of baseline) treat any nausea and vomiting initially with fluids and metaraminol boluses. If systolic BP is above 100mmHg then treat with antiemetics eg. ondansetron 4mg iv slowly or cyclizine 50mg iv (diluted) and check that the surgeon is not applying direct pressure on the stomach, pulling on any bowel / omentum or swabbing out the paracolic gutters.

* *To make up the intrathecal diamorphine dose (if prefilled syringe not available), draw up aseptically 5ml of saline using a filter needle and inject it into a 5mg vial of diamorphine to make up a 1mg/ml solution. Then use a 1 ml syringe and 21G needle to take off 0.3 ml of diamorphine solution (0.3mg) and add it to your volume of 0.5% Hyperbaric bupivacaine.*

## 4. References


1. Cowan CM, Kendall JB, Barclay PM, Wilkes RG. Comparison of intrathecal fentanyl and diamorphine in addition to bupivacaine for caesarean section under spinal anaesthesia**.** *Br J Anaesth.* 2002 Sep;89(3):452-8

2. Saravanan S, Robinson AP, Qayoum Dar A, Columb MO, Lyons GR. Minimum dose of intrathecal diamorphine required to prevent intraoperative supplementation of spinal anaesthesia for Caesarean section. *Br J Anaesth*. 2003 Sep;91(3):368-72

3. Kelly MC, Carabine UA, Mirakhur RK. Intrathecal diamorphine for analgesia after caesarean section. A dose finding study and assessment of side-effects*. Anaesthesia.* 1998 Mar;53(3):231-7.

4. Khaw KS, Wang CC, Ngan Kee WD, Pang CP, Rogers MS. Effects of high inspired oxygen fraction during elective caesarean section under spinal anaesthesia on maternal and fetal oxygenation and lipid peroxidation. *Br J Anaesth*. 2002 Jan;88(1):18-23.

5. Cogliano MS, Graham AC, Clark VA. Supplementary oxygen administration for elective Caesarean section under spinal anaesthesia. *Anaesthesia.* 2002 Jan;57(1):66-9.

6. Khaw KS, Ngan Kee WD, Lee A, Wang CC, Wong AS, Ng F, Rogers MS.Supplementary oxygen for elective Caesarean section under spinal anaesthesia: useful in prolonged uterine incision-to-delivery interval? *Br J Anaesth*. 2004 Apr;92(4):518-522

7. Kinirons B, Mimoz O, Lafendi L, Naas T, Meunier J, Nordmann P. Chlorhexidine versus povidone iodine in preventing colonization of continuous epidural catheters in children: a randomized, controlled trial. *Anesthesiology*. 2001 Feb;94(2):239-44.

8. Rees SG, Thurlow JA, Gardner IC, Scrutton MJ, Kinsella SM. Maternal cardiovascular consequences of positioning after spinal anaesthesia for Caesarean section: left 15 degree table tilt vs. left lateral. *Anaesthesia*. 2002 Jan;57(1):15-20.

9. Kinsella SM.Lateral tilt for pregnant women: why 15 degrees? *Anaesthesia* **58** (9), 835-836.

10. Russell IF. Levels of anaesthesia and intraoperative pain at Caesarean Section under regional block.*International Journal of Anaesthesia* 1995; 4:71

11. Vallejo MC, Mandell GL, Sabo DP, Ramanathan S. Postdural puncture headache: a randomized comparison of five spinal needles in obstetric patients. *Anesth Analg*. 2000 Oct;91(4):916-20.

12. Smith EA, Thorburn J, Duckworth RA, Reid JA. A comparison of 25 G and 27 G Whitacre needles for caesarean section. *Anaesthesia.* 1994 Oct;49(10):859-62.